An 8-bit 56GS/s 64x Time-Interleaved ADC with Bootstrapped Sampler and Class-AB Buffer in 4nm CMOS

A. Serdar Yonar[1,2], Pier Andrea Francese[1], Matthias Brändli[1], Marcel Kossel[1], Thomas Morf[1], Jonathan E. Proesel[5], Sergey Rylov[3], Herschel Ainspan[3], Martin Cochet[3], Zeynep Deniz[3], Timothy Dickson[3], Troy Beukema[3], Christian Baks[3], Michael Beakes[5], John F. Bulzacchelli[3], Young-Ho Choi[4], Byoung-Joo Yoo[4], Hyoungbae Ahn[4], Dong-Hyuk Lim[4], Gunil Kang[4], Sang-Hune Park[4], Mounir Meghelli[3], Hyo-Gyuem Rhew[4], Daniel Friedman[3], Michael Choi[4], Mehmet Soyuer[3], Jongshin Shin[4]

[1]IBM Research Europe - Zurich, Switzerland, [2]ETH Zurich, Switzerland, [3]IBM Research, Yorktown, USA, [4]Samsung Electronics, Hwaseong, Republic of Korea, [5]formerly with IBM T.J. Watson Research Center

yon@zurich.ibm.com

2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits) | 978-1-6654-9772-5/22/$31.00 ©2022 IEEE | DOI: 10.1109/VLSITechnologyandCir46769.2022.9830308

Abstract

A 56 GS/s 8-bit asynchronous SAR ADC fabricated in 4nm CMOS technology is demonstrated. The 16x4 interleaved ADC uses a novel bootstrapping technique and a class-AB follower in the 1st rank interleaver. It achieves a broad input common-mode (CM) range; from 0.3V to 0.6V, the total harmonic distortion stays below -52dB at 4.1 GHz with -0.2dBFS amplitude at 0.8V_{PPD} maximum full scale. The ADC includes analog foreground calibration means for offset, gain, skew, and bandwidth. The measured ENOB is 6.5 at low frequency and stays above 5.2 up to Nyquist frequency. The bandwidth is higher than 27 GHz. The ADC uses a single 0.8V supply voltage and achieves an efficiency of 47 fJ/conv.step.

Keywords: Bootstrapped Track-and-Hold, Class-AB Buffer, Time-Interleaved ADC, asynchronous SAR

Introduction

Modern ADC-based receivers for high-speed serial links rely on time-interleaving to reach required speeds of 112 Gb/s and above. Asynchronous SAR based ADC is a proven efficient architecture that scales well with technology and maintains a good balance between maximum speed and metastability, but requires high interleaving factors to obtain sufficient margins as target sample rates increase [1]. With increased interleaving factors, however, track-and-hold (T&H) performance becomes the main limitation. Using an NMOS switch as the 1st rank sampler followed by a PMOS source follower is a simple and efficient design approach [2] but requires a high supply voltage which is limited by the maximum supported voltage in advanced technology nodes. Moreover, such an approach requires a precisely tuned low input CM voltage, precluding the use of advantageous inverter-based frontend topologies, which demand CM voltages closer to mid-rail [3]. To address these limitations, the reported 8-bit 56GS/s ADC uses a 1st rank T&H circuit comprising a bootstrapped sampling NMOS switch followed by a class-AB complementary buffer. Measurements show that the proposed T&H circuit achieves excellent linearity across a wide input CM range, while using a nominal supply voltage of only 0.8V.

ADC Architecture

The 16x4 interleaved ADC uses a two-rank topology as shown in Fig. 1. The 1st rank consists of sixteen bootstrapped NMOS switches followed by class-AB buffers, each driving four sub-ADCs. Each sub-ADC output is captured by a 256-sample shift register. The ADC receives quadrature clock phases CK4, the PLSMAIN divides CK4 to create 4UI pulses from which the cascaded PLSGEN blocks generate the 2UI tracking and 2UI reset pulses in close proximity to the T&H. The 1st rank T&H must support the full input signal bandwidth and tolerate a wider input CM range with a nominal supply voltage of 0.8V, necessitating a new design approach. The 2nd rank samplers use a conventional bootstrapping technique [4].

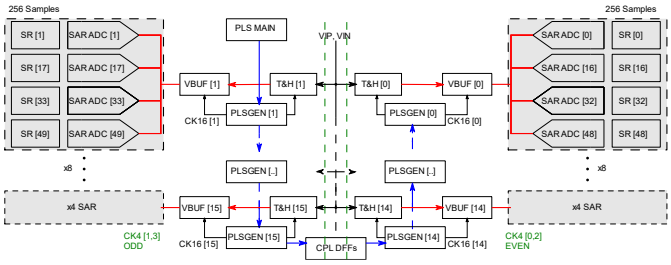


Fig. 1 ADC interleaver architecture and clocking.

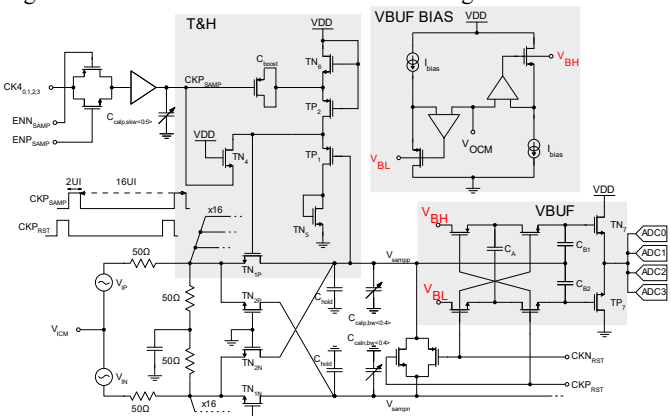


Fig. 2 First rank T&H and class-AB buffer.

1st Rank Bootstrapped Switch

The circuit schematic is shown in Fig. 2. When in track mode, the NMOS switch (TN_{1P}) is on, and the input signal charges the sampling capacitor (C_{hold}) with a linearity error proportional to the on-resistance variation. By applying a constant Vgs, such error can be reduced. This is achieved by the level shifting of a switched PMOS source follower (TP_1), which reaches a higher bandwidth for two reasons. First, contrary to the conventional implementations [4], a constant voltage between the gate of TN_{1P} and the sampling capacitor is maintained, which can be regarded as a constant Vgd bootstrapping. With this arrangement, the input does not see any extra load; rather, the capacitance due to TP_1 becomes part of the sampling capacitor. Second, this topology minimizes the delay between the sampling capacitor and the gate of TN_{1P}, which reduces tracking errors at high frequencies. The PMOS source follower (TP_1) is switched on and off by TP_2 which, ideally, acts as a switched current source when the current flowing out of the C_{boost} capacitor is stable. The size of C_{boost} must be sufficient to limit its discharge during the tracking time, or equivalently the tracking time should be sufficiently short,

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which has the additional advantage of reducing the energy consumed by TP_1. The transition from track to hold mode is, however, critical. When TP_2 is switched off, the TN_{1P} gate is rapidly driven to ground. In this hold state, cross-coupled NMOS switches help improve the high frequency isolation. During the whole operation, all transistors remain below the Vmax specification.

Class-AB Buffer

The class-AB buffer shown in Fig. 2 is used in the 1st rank T&H. Compared to a simple source follower [2], it has both superior linearity and driving capability, and its CM requirements are compatible with inverter-based frontends. Compared to the implementation in [5], the proposed buffer is differently biased; it uses a switched capacitor network to effectively eliminate DC baseline wandering. The bottom plate parasitic capacitors of C_{B1} and C_{B2} are embedded in the sampling capacitors like the bootstrapped switch parasitics. During the reset phase, the T&H CM output voltage is shifted by C_{B1} and C_{B2} to the values defined by the replica bias circuit.

Asynchronous SAR Sub-ADC

Speed enhancement techniques such as loop unrolling [2] are not beneficial in the 4nm node where the CDAC wiring is the main speed bottleneck. The single loop SAR architecture has the most compact floorplan and only one analog offset to calibrate. Fig. 3 shows the sub-ADC architecture. The StrongARM comparator clock generation is concurrent with the memory capture and the CDAC settling. After the synchronous reset (RES), each differential memory cell is enabled and disabled (ENB_REG) in an asynchronous domino sequence. Comparator decisions are latched in a self-timed manner and held raising the KEEP signal. Two resistive ladders are employed for offset and gain correction.

Measurement Results

The ADC test chip was fabricated in Samsung 4nm FinFet technology. Fig. 5 shows the die photo and the ADC layout details. The active circuit occupies 0.078 mm^2. The ADC test chip was placed on a GL102F 4-2-4 substrate and mounted to a Nelco4000-13 test card for characterization. Reported measurement results were taken without any post-processing after analog foreground calibrations were performed and stored on chip. A single 0.8V supply and 0.8V_{PPD}FS input at 56 GS/s were used during these measurements. At f_{sine}=4.1 GHz, the ENOB is 6.5, SFDR is 61.9dBc and INL stays within -1.14 and 1.13 LSBs, as shown in Fig. 4. To showcase the proposed bootstrapping technique, we swept the input CM from 0.3V to 0.6V; the reported THD in Fig. 4(f) remains below -52dB at 4.1 GHz. The ENOB at Nyquist is 5.2 and the 3dB-bandwidth is higher than 27 GHz. The power consumption is 240mW, (40mW clock receiver and buffers, 43mW interleaver, 157mW ADCs) which corresponds to 47 fJ/conv.step. These results are on par with the best published numbers and confirm the effectiveness of the proposed bootstrapping technique.

Conclusion

This work demonstrates a fast 1st rank bootstrapped sampler with excellent linearity operating within a wide input CM voltage range from the lowest voltage supply. The circuit is implemented in 4nm CMOS technology and showcased in an 8-bit ADC at 56 GS/s with state-of-the-art performance as reported in the Table I.

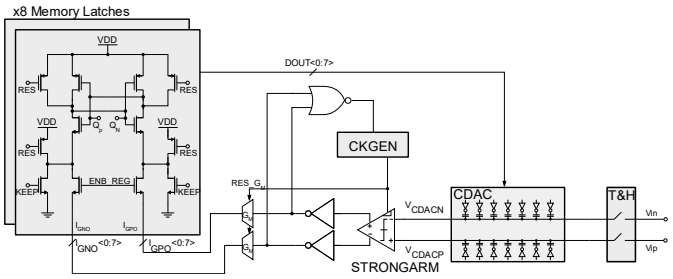


Fig. 3 Sub-ADC architecture.

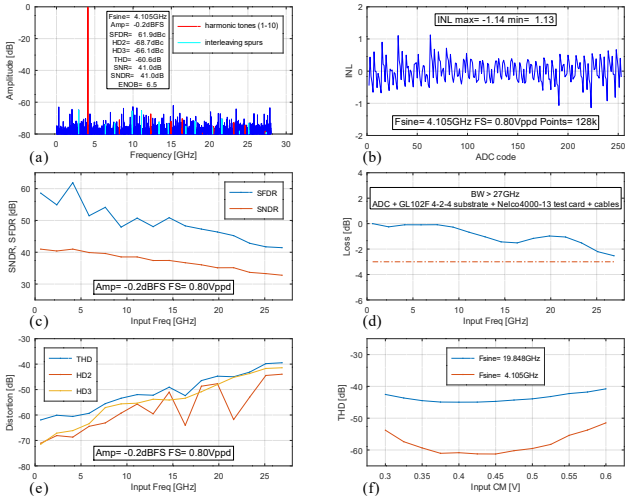


Fig. 4 (a) Output spectrum at f_{sine} = 4.1 GHz (b) INL distribution (c) SFDR, SNDR vs. f_{sine} (d) Loss vs. f_{sine} (e) THD, HD2, HD3 vs. f_{sine} (f) THD vs. V_{cm} at 19.8 GHz and 4.1 GHz.

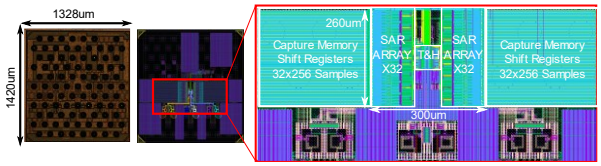


Fig. 5 ADC die micrograph and layout details.

TABLE I: SUMMARY AND COMPARISON OF RESULTS

	[8]*	[7]*	[2]	[6]	This Work
Technology	7nm	20nm	14nm	28nm	4nm
Resolution (bits)	8	8	8	8	8
fs (GS/s)	97	64	72	56	56
SNDR @ fin,low dB @ GHz	41@DC	37.6@8	39.3@2	40.5@7	41.0@4.1
SNDR @ fin,high dB @ GHz	32@36	33.7@16	32.7@27.5	33@27	32.8@26.9
INL (min,max)	NA	NA	(-2.37, 4.62)	NA	(-1.14,1.13)
BW (GHz)	>40	20	21	31.5	>27
Power (mW)	305	950	235	702	240
FOM @ fin,low (fJ/conv.step)	34	240	43	145	47
FOM @ fin,high (fJ/conv.step)	96	375	121	344	117
Supply (V)	1st rank@1.2 2nd rank@0.9	1.5, 1	sub-ADC@0.8 Inerleaver@0.9	sub-ADC@0.95, Input Buffers @0.95/-0.9	0.8
Area (mm^2)	NA	~1.7	0.15 (active)	0.878 (active)	0.078(active)

*Includes AFE

Acknowledgements The authors thank the Samsung Foundry IP Development Team, and the Samsung Semiconductor India Research layout team.

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